an operating system itself runs programs and it can't just run another set of programs which manage the computer (another operating system). As we've already mentioned, this is done using virtualization software.

Virtualization software is a program which can fool an operating system into believing that it is utilising the computer's hardware which isn't the case. The software actually creates an environment wherein it emulates the various hardware components of the computer. When one tries to launch (or install) an operating system, the virtualization software will inform the operating system being run about the hardware that it emulated rather than about the actual hardware installed on the computer. This technique is called "sandboxing".

The operating system is run inside a sandbox (or a virtual environment) which looks real but isn't. When creating a new Virtual Machine, you need to tell the virtualization software some information such as:

- The amount of RAM you want to allocate to that operating system: This amount of memory is reserved by the software from the real RAM on the system.
- The number of cores you want to allot the operating system: Some virtualization software allow you to allot more than the actual

CAUTION

Since the amount you allot for the hard disk is allocated inside a file (which will be kept on your real system), you're limited by the free space available on the partition where you're saving the virtual hard disk file. So if you're trying to allocate 20 GB of hard disk space to the virtual machine and you only have 10 GB of space on the partition where the virtual hard disk file will be saved, this space will fall short and either the hard disk wouldn't be created, or if it gets created, you'll run into problems like slow operation or crashes. Remember that in such situations, the behavior would depend on the nature of your virtualization software. When the space gets filled up, one thing is sure – your real system (the one where the virtualization software is installed) will begin to crawl. Moreover, (as we already said) depending on the virtualization software you're using, the virtual machine may crash and your whole system might hang or crash. So ensure you have enough free space on your hard disk to create the amount of space you want the virtual machine to have.